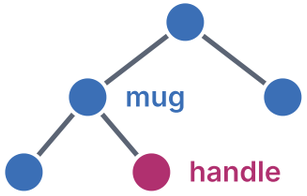
Containment tree



$$\text{parent}(v) = \arg\min_u |m_u|$$
$$\text{s.t. } |m_u \cap m_v| \geq 0.8\,|m_v|$$

Answer

$m_{v^\star}$

one mask



selection scores over the pool



Selection head

$$s_v = \lambda \cos\big(\psi_q(h_{\langle \text{SEL} \rangle}),\, \psi_k(z_v)\big) + \beta$$
$$v^\star = \arg\max_v s_v$$

MLP ψ_q $h_{\langle \text{SEL} \rangle}$ at layer $\ell^\star$ z_v MLP ψ_k

Qwen3-VL-4B + LoRA

one pass over image, query and mask tokens; LoRA rank r = 16

e_j <SEL> $\hat{x}_v$

Vision encoder | **Text tokenizer** | **Mask tokenizer**

Inputs



Image

PACO test

Query

"Locate the handle of the mug."

phrases $\phi(x)$

the handle of the mug | the mug

Candidates from SAM 3 (frozen)



N = 260: point grid + text phrases (#rank)

Mask tokenizer one token per candidate

candidate



m_v

Appearance $a_v = \sum_j \tilde{c}_{v,j}\, e_j$

MLLM image tokens in the mask

Geometry $g_v \in \mathbb{R}^9$

box, area, depth in the tree

Source $q_v \in \mathbb{R}^5$

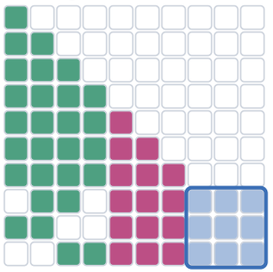
grid or SAM 3 text: rank, score

concatenate

MLP + RMS norm

$$\hat{x}_v = \text{MLP}\big([\,a_v;\, g_v;\, q_v\,]\big)$$ position: mask centroid

Region-restricted attention every layer



- a mask token sees only its own region and the query
- image and query tokens never see the mask tokens
- between masks: learned bias $b_{\ell,\,\text{rel}(u,v)}$ by tree relation